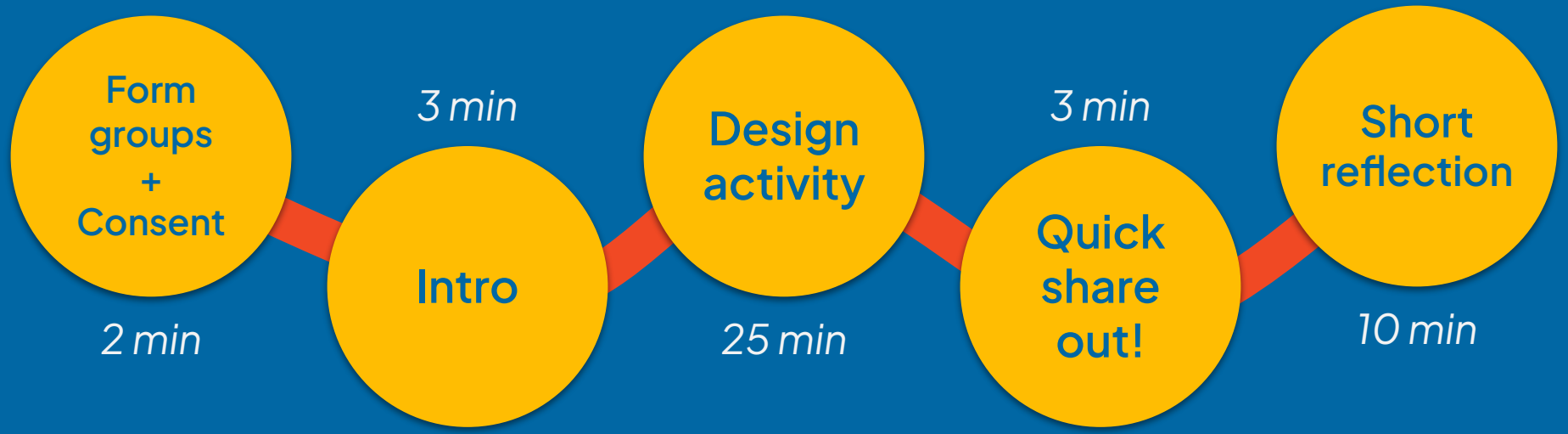
Robot Design Workshop!

Today's Plan

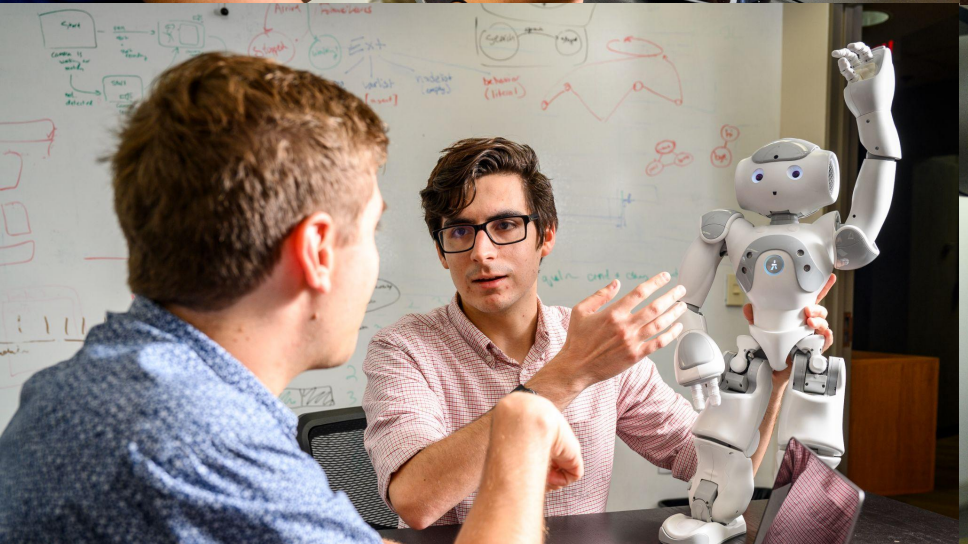

Form groups + Consent
2 min
3 min
Intro
Design activity
25 min
3 min
Quick share out!
Short reflection
10 min

What will we do today?

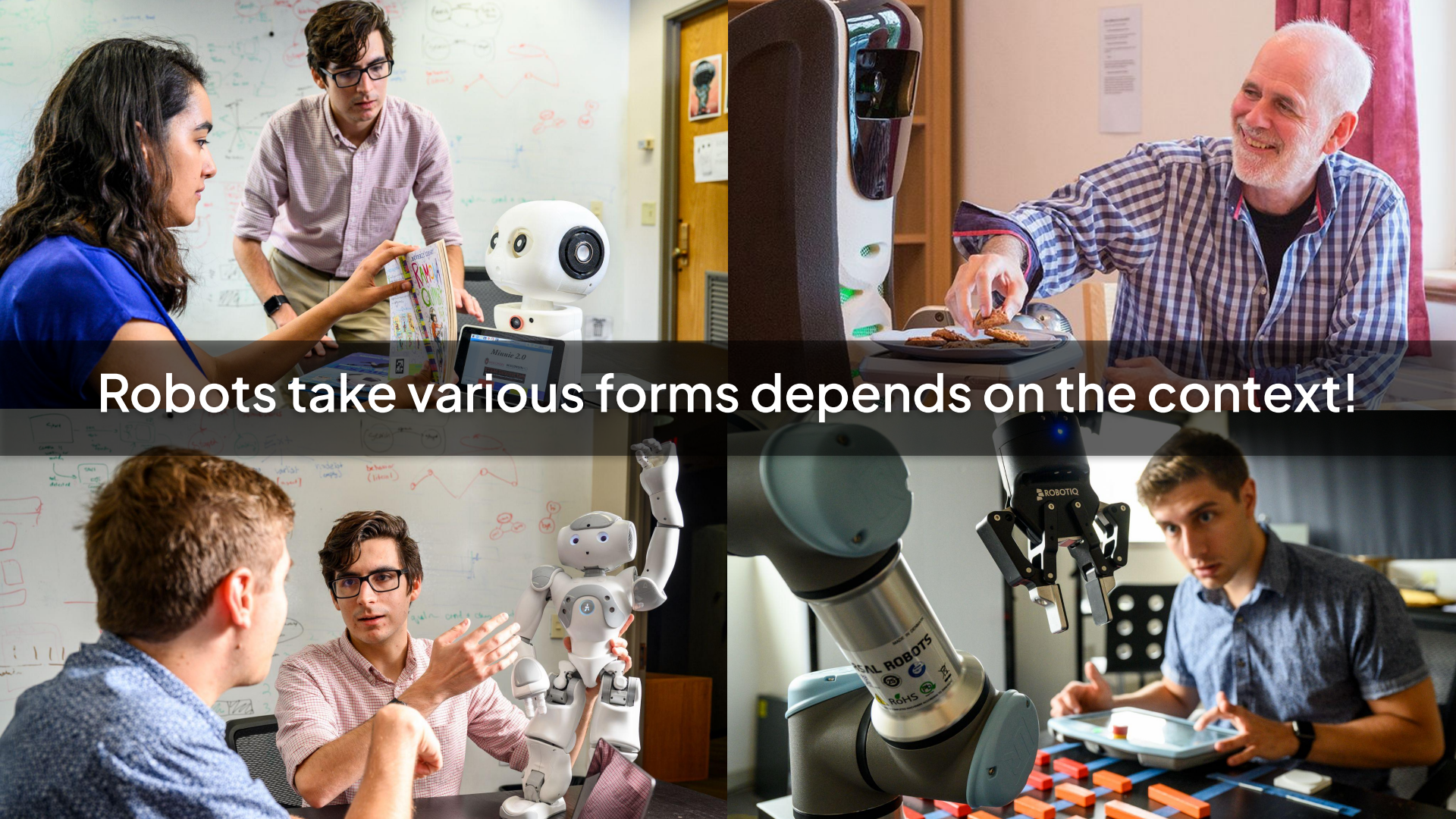
We will *design* a robot!

Oh? *Robot*? 🤖

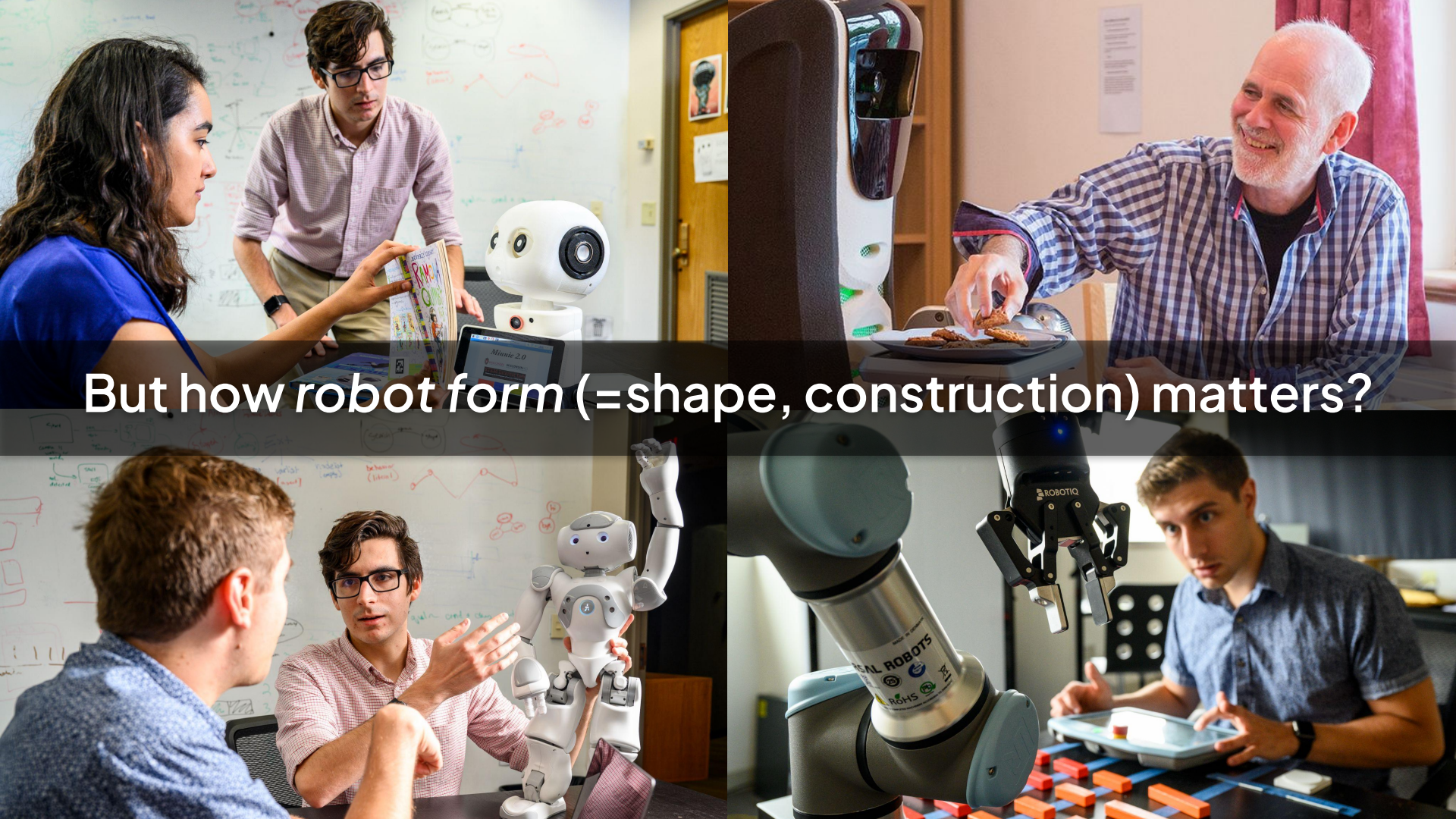
Minnie 2.0



ROBOTIQ
ROHS



Robots take various forms depends on the context!



But how *robot form* (=shape, construction) matters?

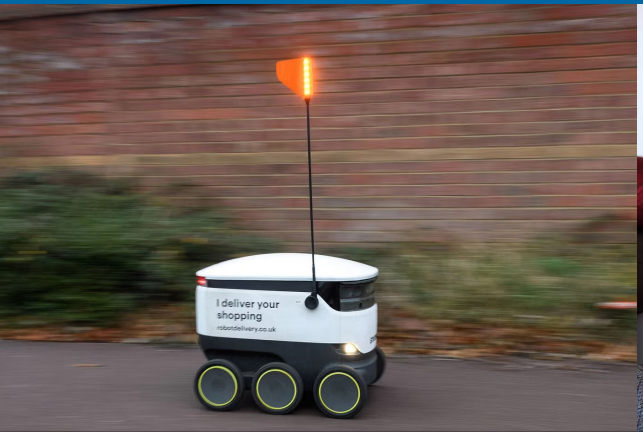
enable locomotion...

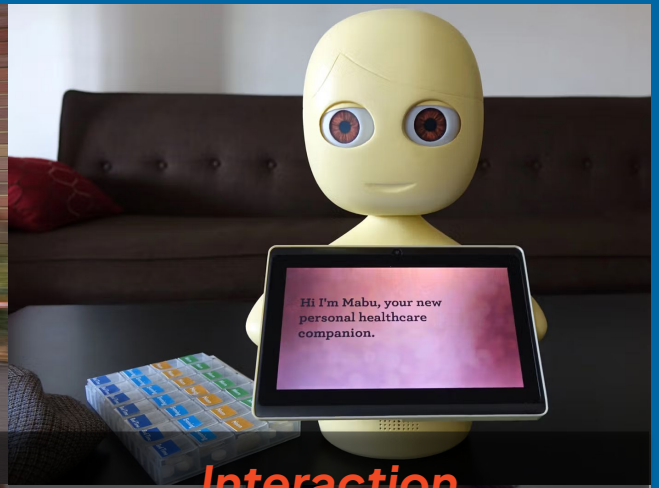
Function

enable movement

enable locomotion...

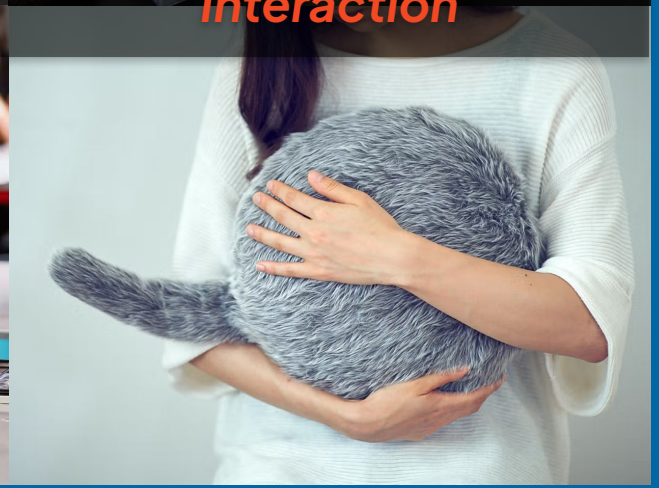
give focus of attention...





Function

Interaction



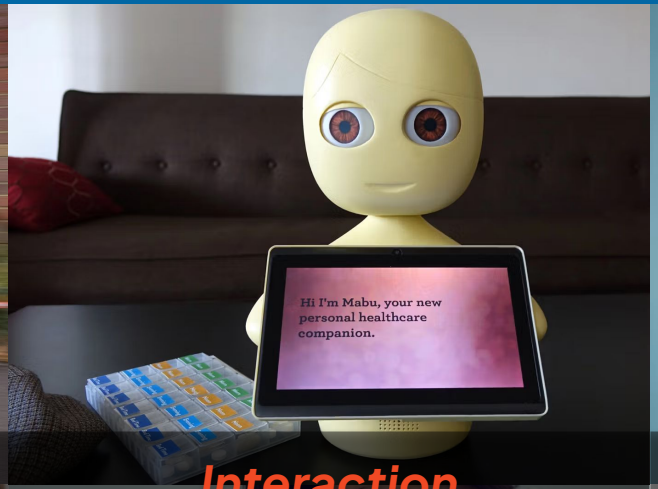
enable movement

imply proximity...

enable locomotion...

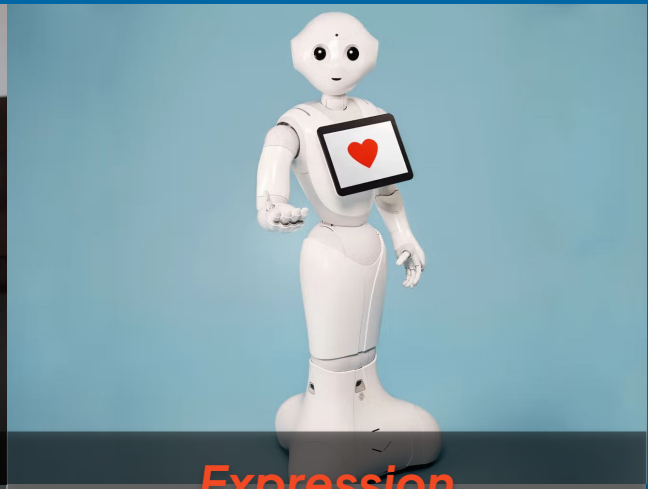
Function

give focus of attention...



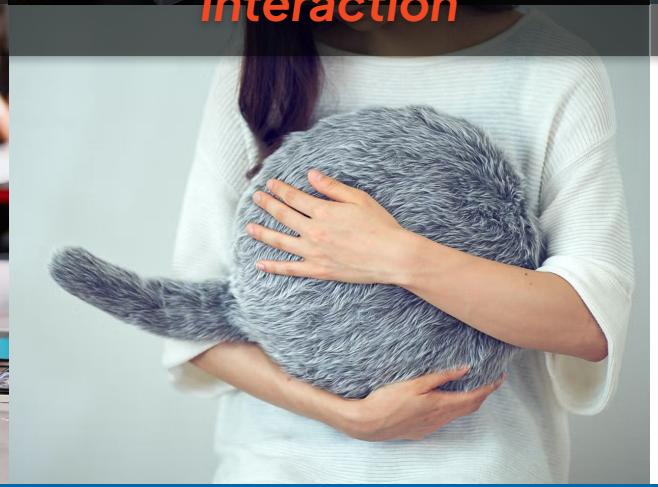
Interaction

create impression...



Expression

enable movement



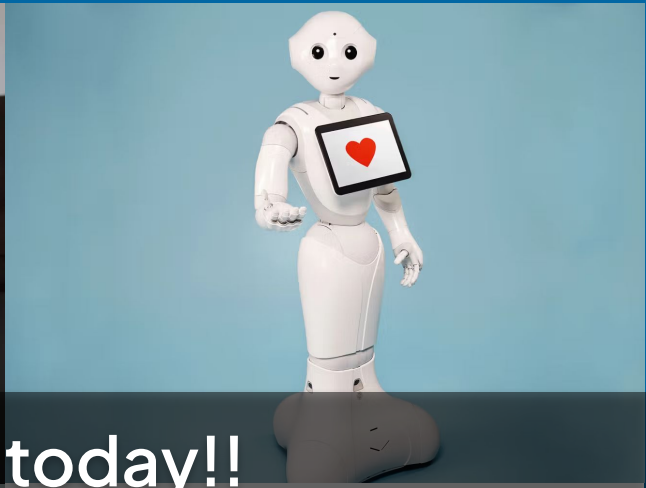
imply proximity...

convey its presence...

enable locomotion...

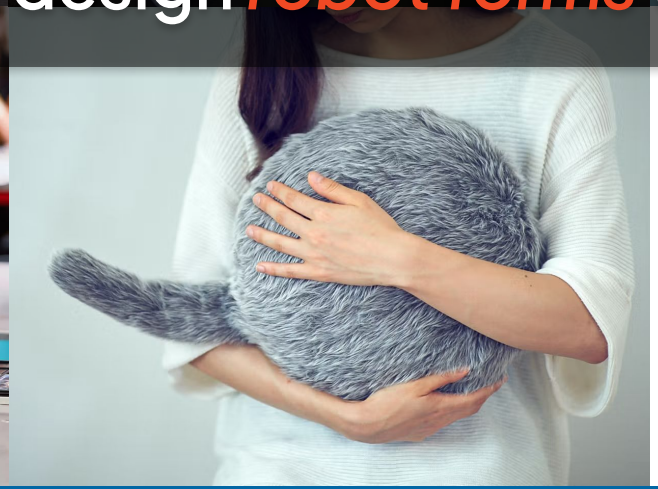
give locus of attention...

create impression...



We will design *robot forms* today!!

enable movement



imply proximity...

convey its presence...

Your Turn!

Step 1

Concept Making

Brainstorm, sketching, etc.

Step 2

Form Exploration

Use blocks!!

Step 3

Finalize

Finalize your design!

Step 1: Concept Making!

3 mins

1. Pick one context from the list below that interests you
2. Use the worksheet to jot down ideas or make sketches, think about what the robot might be like…

A. **Design a patrol robot that can navigate a crowded public space.**

B. **Design a delivery robot that can handle fragile objects in the outdoor environment.**

C. **Design an education assistant robot that can support learning in a classroom environment.**

Step 2-1: Pick Keywords

1 mins

1. Refer back to your concept from the previous step
2. Pick one keyword card from the list that best connect to your concept (or that just caught your eyes!)

Step 2-2: Form Exploration!

15 mins

1. Familiarize yourself with the blocks!
2. Design one robot for each keyword using the blocks. (You will build 2 robots in total)
3. Use blocks as many(few) as you want. + Feel free to use clay if you want to create any elements/components.
4. Let the facilitator know when you feel done! Will take photos.

Step 3: Finalizing!

5 mins

1. Pick one robot you designed in the previous step and finalize it by answering the following questions. Use worksheets!
2. Questions:
   a. *What is the name of the robot?*
   b. *What would the size, color, texture or material of the robot be ?*
   c. *How would the robot behave? (e.g., how would the robot interact with people or environment?)*

Quick Share-Out!

3 mins

1. Share the robot you have designed to other groups!
2. Tell everyone about:
   a. Its name
   b. Its main characteristics
   c. How it behaves or interacts
   d. Your favorite thing about it